SER216 – Software Quality Assurance

# Final Practical Project

# Digital Parking Permit System

Student: **Rahmatulloh Toirov**
Repository: **SER216_Final_Project_RahmatullohToirov**
Branch: **feature/upload-project-files**

GitHub Repository: https://github.com/RahmatullohToirov/SER216_Final_Project_RahmatullohToirov

# Table of Contents

# Task 1 – Requirements

The following requirements have been identified for the Digital Parking Permit System. Functional requirements describe what the system must do, while non-functional requirements describe how the system must perform.

## Functional Requirements

- **FR-1 Vehicle Registration:** The system shall allow students and staff to register their vehicles by providing a valid university ID and vehicle details (make, model, year, license plate).
- **FR-2 Parking Permit Request:** The system shall allow registered users to request a parking permit for a specified parking zone and time duration. Users must be logged in to request a permit.
- **FR-3 Zone Availability Check:** The system shall display the available parking zones and their current real-time occupancy status so that users can choose an appropriate zone before requesting a permit.
- **FR-4 Permit Verification:** The system shall allow security staff to verify the validity of any parking permit by entering or scanning the unique permit ID, displaying the owner name, vehicle details, zone, and expiry date.

## Non-Functional Requirements

- **NFR-1 Performance:** The system shall process permit verification requests within 2 seconds under normal operating load (up to 500 concurrent users).
- **NFR-2 Availability:** The system shall maintain 99.9% uptime during all university operating hours (Monday–Friday, 07:00–22:00).

# Task 2 – Workflow Diagram

The workflow below illustrates the end-to-end process for a user obtaining a parking permit and having it verified by security staff. The five stages are sequential and each stage must be successfully completed before the next begins.

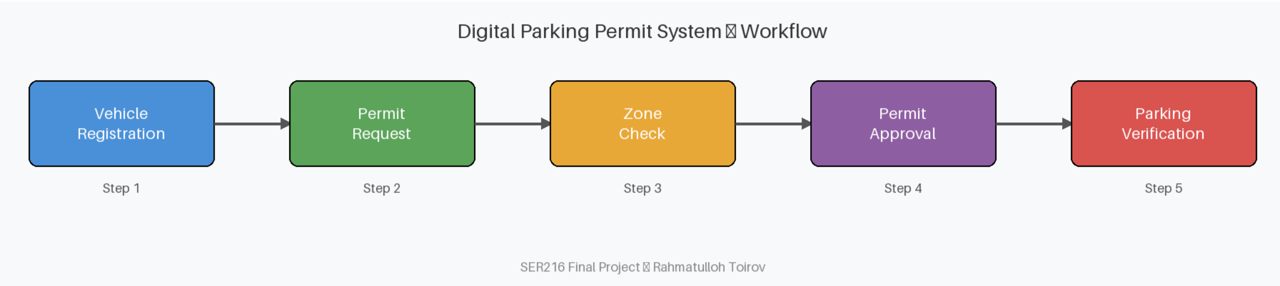


*Figure 1: Digital Parking Permit System Workflow*

## Stage Descriptions

**Step 1 – Vehicle Registration:** The user (student or staff) submits their university ID and vehicle information to the system. The system validates the details and creates a user profile linked to the vehicle.

**Step 2 – Permit Request:** The registered user selects a desired parking zone and permit duration and submits a permit application. The system checks that the user has no outstanding violations before proceeding.

**Step 3 – Zone Check:** The system checks the real-time availability of the requested zone. If the zone is at full capacity, the user is notified and may select an alternative zone.

**Step 4 – Permit Approval:** The system (or an administrator) reviews and approves the permit request. A unique permit ID is generated and issued digitally to the user.

**Step 5 – Parking Verification:** When the user parks, security staff verify the permit by scanning or entering the permit ID. The system confirms validity, zone, and expiry date in real time.

# Task 3 – Test Case Table

The following six test cases cover three test types: System Testing (checks the complete integrated system), Acceptance Testing (checks whether the system meets user/business expectations), and Regression Testing (ensures previously fixed issues have not re-appeared after changes).

| Test ID | Feature | Input / Data | Expected Result | Test Type |
|---|---|---|---|---|
| TC-001 | Vehicle Registration | Student ID: S12345<br>Vehicle: Toyota Camry 2020<br>Plate: ABC-1234 | Vehicle successfully registered; confirmation message shown with unique Registration ID (e.g., REG-00123) and timestamp | System Test |
| TC-002 | Permit Verification | Permit ID: P98765<br>Verified by security staff | System displays permit as VALID with owner name, zone, expiry date | System Test |
| TC-003 | Parking Permit Request | Registered user requests permit for Zone A, duration: 1 semester | Permit approved; user receives digital permit with unique ID | Acceptance Test |
| TC-004 | Zone Availability Check | User selects Zone B to check availability | System shows current available spaces in Zone B in real time | Acceptance Test |
| TC-005 | Expired Permit Re-verification | Test setup: create permit P11111 with expiry = yesterday's date (system date minus 1 day).<br>Scan permit P11111 | System shows permit P11111 as EXPIRED and denies parking access | Regression Test |
| TC-006 | Vehicle Re-registration After Update | User updates plate from ABC-1234 to XYZ-5678 and requests new permit | New permit issued with updated plate; old permit invalidated | Regression Test |

*Table 1: Test Cases for the Digital Parking Permit System*

## Task 4 – Defect Analysis

**Defect Description:** "A valid parking permit is shown as expired during security verification."

| Item | Student Answer |
|---|---|
| **Defect Category** | Logic / Data Synchronization Error |
| **Severity** | High – prevents security staff from correctly verifying valid permits, blocking legitimate users from parking |
| **Priority** | High – must be fixed before the next release as it directly impacts core system functionality and user expe |
| **Possible Root Cause** | The permit expiry date/time is not being read or compared correctly in the verification module. Likely caus<br>• Timezone mismatch between the server and client<br>• Date format inconsistency (e.g., DD/MM vs MM/DD)<br>• Off-by-one error in the expiry date calculation (e.g., permit expires at end-of-day but system marks it exp |
| **Regression Test** | After the fix is applied:<br>1. Create a permit with today's date as the expiry date – verify it shows VALID<br>2. Create a permit expiring tomorrow – verify it shows VALID<br>3. Retrieve a permit that expired yesterday – verify it shows EXPIRED<br>4. Test across different timezones to ensure consistency |

*Table 2: Defect Analysis – Valid Permit Shown as Expired*

## Task 5 – GitHub Evidence

All required GitHub activities have been completed for this project. The following checklist documents each completed task along with the relevant evidence available in the repository.

**■ Repository Created**

- Repository name: SER216_Final_Project_RahmatullohToirov
- URL: https://github.com/RahmatullohToirov/SER216_Final_Project_RahmatullohToirov

**■ README.md Added**

- The README.md file describes the project, lists tools used, explains the repository structure, and summarises all five tasks.

**■ Project Files Uploaded**

- All required files have been uploaded:
- • final_report.pdf – complete project report
- • diagrams/workflow_diagram.png – workflow diagram
- • tables/test_cases.csv – test case table
- • notes/tools_used.md – documentation of tools

**■ Feature Branch Created**

- Branch name: feature/upload-project-files
- Created from main and used for all project file uploads.

**■ At Least 3 Meaningful Commits Made**

- Commit history includes:
- 1. Add project report structure
- 2. Add workflow diagram and test case table
- 3. Add defect analysis and update README
- 4. Add final PDF and complete project files

**■ Pull Request Created**

- Pull request created from feature/upload-project-files → main, describing all changes and linking to project tasks.

**■ Pull Request Merged**

- Pull request reviewed and merged into main branch after verifying all content.

**■ Final PDF Uploaded**

- final_report.pdf has been uploaded to the root of the repository.

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**Required GitHub Evidence Screenshots** are stored in the *screenshots/* directory of this repository, including: repository homepage, README.md view, commit history, feature branch, pull request, and merged pull request.

## Summary

This report documents the complete analysis and planning deliverables for the Digital Parking Permit System as required by the SER216 Final Practical Project. The five tasks covered in this report are:

- **Task 1 – Requirements:** Four functional requirements (vehicle registration, permit request, zone check, permit verification) and two non-functional requirements (performance, availability) were defined.
- **Task 2 – Workflow Diagram:** A five-stage workflow was designed and illustrated: Vehicle Registration → Permit Request → Zone Check → Permit Approval → Parking Verification.
- **Task 3 – Test Cases:** Six test cases were created covering two system tests, two acceptance tests, and two regression tests for key features of the parking permit system.
- **Task 4 – Defect Analysis:** The defect 'A valid parking permit is shown as expired during security verification' was analysed. It was classified as a High severity/priority Logic/Data Synchronization Error, with likely root causes of timezone mismatch or date format inconsistency.
- **Task 5 – GitHub Evidence:** All required GitHub tasks were completed including repository creation, README, feature branch, 3+ commits, pull request, merge, and PDF upload.

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